Non-contact Measurement of Cable Profile using Computer Vision

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***Abstract*—Accurate sag estimation of lines is essential for ensuring safety and structural reliability. Traditional sag measurement methods require manual inspection or expensive sensors. This project presents an automated image-processing-based approach for sag estimation using overhead cable images. The method involves edge detection, morphological processing, skeleton extraction, and curve fitting using the catenary equation. The fitted curve is then used to calculate sag values and evaluate fitting accuracy. Experimental results demonstrate reliable detection of cable profiles and successful curve approximation using real-world image data. The proposed system offers a low-cost and automated solution suitable for remote monitoring applications.**

***Index Terms*—Sag Estimation, Catenary Curve, Image Processing, Skeletonization, Curve Fitting, Computer Vision**

I. Introduction

Overhead transmission lines naturally form a curve due to gravity and tension forces acting on the cable. This curve, known as a catenary curve, determines the sag of the transmission line. Sag monitoring is critical to prevent mechanical failure, excessive tension, and safety hazards.

Traditional sag measurement techniques involve manual inspection or installation of dedicated sensors, which increases operational cost and maintenance effort. With the advancement of computer vision and image processing techniques, it is now possible to automatically detect cable geometry from images and compute sag values without direct physical measurement.

This project proposes an automated system that processes cable images, extracts the cable structure, fits a mathematical catenary model, and calculates sag parameters accurately.

II. Methodology

The proposed method consists of multiple stages, beginning from image acquisition to sag calculation.

A. Image Acquisition

Images of overhead transmission lines are collected from publicly available datasets. These images contain suspended cables forming curved shapes suitable for analysis.

B. Grayscale Conversion

The input RGB images are converted into grayscale images to simplify processing and reduce computational complexity.

C. Edge Detection

Canny edge detection is applied to identify cable boundaries. This step extracts high-contrast edges corresponding to the cable structure.

D. Morphological Processing

Morphological closing is performed to fill small gaps in detected edges and improve cable continuity.

E. Skeletonization

The processed binary image is skeletonized to obtain a single-pixel-wide representation of the cable. This reduces redundancy and provides accurate coordinate points for curve fitting.

F. Catenary Curve Fitting

The cable shape is modeled using the catenary equation:

$$y = a \cosh\left(\frac{x - x_0}{a}\right) + y_0 \tag{1}$$

Curve fitting is performed using nonlinear optimization techniques to estimate the parameters a, x_0, and y_0.

G. Sag Calculation

Sag is computed as the vertical difference between the lowest point of the cable and the highest support points derived from the fitted curve.

H. Error Evaluation

Mean absolute error between detected skeleton points and fitted curve points is calculated to evaluate model accuracy.

III. Results

The developed system successfully detected cable structures from input images and fitted catenary curves accurately. The final outputs include:

- Edge-detected cable images
- Skeletonized cable structures
- Fitted catenary curve visualization
- Sag calculation values
- Error measurement table

Typical outputs include visual overlays of detected cable points along with fitted curves. The results demonstrate that the proposed method provides consistent sag estimation across multiple images.

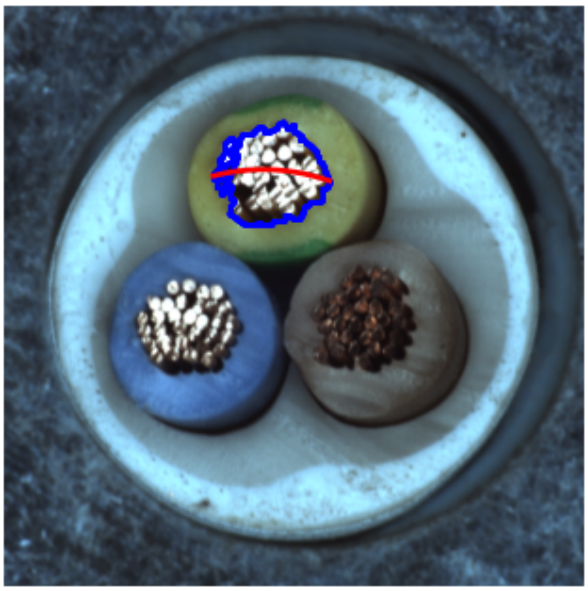

Fig. 1. Catenary Curve Fitting on Cable Image

IV. Discussion

The proposed approach successfully models overhead cable shapes using mathematical curve fitting. Skeletonization plays a critical role in reducing noise and improving accuracy. The fitting results depend on image quality and edge clarity. Images with excessive noise or occlusion may lead to reduced accuracy.

However, the system provides an efficient and scalable solution suitable for automated inspection environments such as drone-based monitoring.

V. Conclusion

This project presents an automated sag estimation system using image processing and curve fitting techniques. The method effectively detects cable shapes and fits mathematical models to compute sag values. The approach reduces dependency on manual measurements and enables scalable infrastructure monitoring. Future improvements may include integration with drone systems and real-time processing pipelines.

VI. Future Work

Future enhancements to the system may include:

- Real-time video-based sag estimation
- Deep learning-based cable detection
- Integration with UAV inspection systems
- 3D sag estimation using stereo imaging

References

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